Clodronate

Clodronate is a cancer drug treatment used to treat and prevent problems caused by the effect of cancer on the bones.

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What is clodronate?

Clodronate belongs to a group of drugs called bisphosphonates. It can be used to treat:

- high levels of calcium in the blood caused by cancer that has spread to the bones from a primary cancer somewhere else in the body (secondary bone cancer)
- bone weakness or pain caused by myeloma or breast cancer that has spread to the bones.

Clodronate can be given with other cancer treatments.

It is best to read this information with our general information about cancer treatments and the type of cancer you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree (consent) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the electronic Medicines Compendium (eMC) website, which has patient information leaflets (PIL) for individual drugs.

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The effect of cancer on the bones

Bone cells constantly renew themselves. This helps bones keep their strength and shape.

There are 2 types of cell inside the bones. These are called:

- osteoclasts, which break down and remove old bone
- osteoblasts, which build new bone.

As we get older, osteoclasts begin to break down more bone than osteoblasts make to replace it. This means our bones slowly become thinner (less dense). In some people, too much of the bone density is lost and this increases the risk of osteoporosis and bone fractures.

Some secondary bone cancers make a substance that causes osteoclasts to destroy more bone than can be replaced. The affected bone becomes weak and painful and can break more easily.

Bones contain calcium, which gives them strength. A bone affected by cancer may cause calcium from the bones to be released into the blood. A raised level of calcium in the blood is called hypercalcaemia. This may cause symptoms including:

- feeling sick (nausea)
- being sick (vomiting)
- feeling thirsty
- needing to pee (pass urine) more often
- constipation
- tiredness
- irritability
- confusion.

How clodronate works

Clodronate reduces the activity of osteoclasts. This can help reduce pain and strengthen the bone.

Clodronate also reduces the amount of calcium lost from the bones. This helps calcium levels in the blood return to normal.

How clodronate is given

During your course of treatment, you will meet with someone from your cancer team, such as a:

- cancer doctor
- chemotherapy nurse or specialist nurse

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- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have clodronate tablets.

Clodronate comes in tablets. This means you can take it at home. You may have it with other cancer drugs. You usually take clodronate once or twice a day.

Taking your clodronate tablets

Clodronate should be swallowed whole with water. Never take it with milk as this reduces the amount of clodronate your body can absorb. Do not chew or crush the clodronate tablets.

You need to drink plenty of fluids, such as water, while you are taking clodronate treatment.

Always take clodronate tablets exactly as explained. This is important to make sure they work as well as possible for you.

If you take clodronate once a day, take your tablets:

- in the morning
- on an empty stomach
- with a glass of water.

After taking clodronate, do not eat, drink (other than water) or take any other medications by mouth for 1 hour.

If you take clodronate twice a day, take the first dose as explained earlier. Take the second dose between meals. Take it at least 2 hours after eating, drinking (other than water) or taking any other medications by mouth.

You should then wait at least 1 hour before eating, drinking (other than water) or taking any other medications by mouth.

Other things to remember about your tablets:

- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
- If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.
- If you take more clodronate than you should, contact the hospital straight away on the 24-hour number you have been given for advice. Drink plenty of water.
- Do not throw away unused tablets. Return them to your cancer team at the hospital.
- After taking the clodronate tablets, avoid lying down for a while afterwards. You can sit or stand. This is to avoid you getting tummy pain.

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How long clodronate is given for

You usually take clodronate every day. Your cancer team will discuss your treatment plan with you. They may give you a copy of the treatment plan to take home.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Dental check-up before treatment

Rarely, this treatment can affect the jawbone. You can read more about this in the side effects section below.

Keeping to a regular oral hygiene routine keeps your mouth and teeth healthy. This reduces your risk of jaw problems. You usually need a dental check-up to check for any problems before you start treatment.

If you need urgent treatment to lower high calcium levels in your blood, you do not usually need to have a dental check-up first.

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Side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Feeling sick

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

If you feel sick, take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

Diarrhoea

This treatment may cause diarrhoea. Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Sore throat or indigestion

This may be a sign that the drug is irritating your throat or gullet (the tube between the mouth and the stomach). Tell your doctor before having any more clodronate if:

- swallowing is painful or difficult
- you have indigestion that is new or getting worse.

Numbness or tingling

You may notice numbness or tingling around the mouth, or in the fingers and toes. Contact your doctor or the hospital straight away if you have these symptoms. This may be caused by low levels of calcium in your blood. You will have regular blood tests to check your calcium levels.

Your doctor may ask you to take calcium and vitamin D supplements, unless you are having this treatment to lower the levels of calcium in your blood. Your doctor will let you know if you need any supplements.

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Effects on the liver and kidneys

This treatment can affect how your kidneys or liver work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys and liver are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids also helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Pain in the thigh, hip or groin

Rarely, people having this treatment develop a break (fracture) in their thigh bone without an obvious cause. Sometimes both thigh bones are affected.

If you have any thigh, hip or groin pain, tell your doctor. Let them know that you are taking bisphosphonates. They can arrange tests to check the thigh bones for any signs of bone weakness or fracture.

Jaw problems

Rarely, this treatment can affect the jawbone. Healthy bone in the jaw becomes damaged and dies. This is called osteonecrosis of the jaw (ONJ). It can cause:

- pain
- loosening of the teeth
- problems with the way the gums heal.

The risk of jaw problems is higher after some types of dental treatment. It is also higher in people who smoke, have gum disease or dentures that do not fit well.

It is important to avoid having dental treatment that could affect your jawbone when you are having bisphosphonates. This includes having a tooth or root removed or dental implants put in. You can still have fillings, gum treatments or a scale and polish.

To reduce your risk of developing jaw problems, your doctor will advise you to:

- have a full dental check-up before starting treatment
- look after your teeth and gums during treatment (ask your dentist for advice)
- tell your dentist you are taking a bisphosphonate before having any dental treatment.

Tell your cancer specialist and dentist straight away if you at any time you develop:

- pain
- swelling or redness in your gums
- numbness or heaviness in your jaw
- loose teeth.

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Ear problems

Very rarely, bones in the outer ear may be affected by this treatment. Always tell your doctor if you have any:

- ear pain
- discharge from your ear
- ear infections.

You should also let them know if you notice any other changes in your ears or hearing.

Allergic reaction

Rarely, this treatment can cause an allergic reaction. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

If you develop any of these signs or feel unwell, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Effects on the eyes

Your eyes may become watery and feel sore. Your doctor, nurse or pharmacist can give you advice and eye drops to help with this.

Contact the hospital on the 24-hour number if:

- your eyes get red and inflamed (conjunctivitis)
- you have pain in your eyes
- you notice any change in your vision.

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Other important information

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- herbal drugs and complementary or homeopathic therapies
- recreational drugs – for example, cannabis.

Lactose

This treatment may contain lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and coronavirus (covid). They may also recommend other vaccines, such as Shingrix® for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Contraception

Your doctor, nurse or pharmacist will advise you not to get pregnant or make someone pregnant while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to breastfeed while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

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Fertility

Some cancer drugs can affect whether you can get pregnant or make someone pregnant. If you are worried about this, it is important to talk with your doctor before you start treatment.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Visit the electronic Medicines Compendium (eMC) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

Cancer Information Development team

Author

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Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

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- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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